🚀 End-to-End Deployment of My Portfolio (Dev → UAT → Prod)

🔷 1. Built the Portfolio Application

- Created a **static website (HTML, CSS, JS)**
- Included:
  - Resume
  - Profile image
  - Projects & experience

👉 This forms the **application layer**

🔷 2. Version Control (GitHub)

- Created a repository to manage code and changes

  👉 Repo: https://github.com/manojp-03/portfolio-website

🔷 3. Development Environment (Local)

- Tested the application locally before deployment

  👉 URL: http://localhost:5050/

👉 This was my **Dev environment**

🔷 4. UAT / Staging Deployment (Firebase Hosting)

- Deployed using **Firebase Hosting (GCP native service)**
- Benefits:
  - Global CDN
  - Automatic HTTPS
  - Fast static hosting

  👉 UAT URL: https://portfolio-website-2096a.web.app

👉 This acted as my **Staging / UAT environment**

🔷 5. Created GCP Infrastructure (Compute Engine VM)

- Provisioned a **GCP VM**

👉 This is the **Infrastructure layer (IaaS)**

🔷 **6. Installed & Configured Nginx**

- Installed **Nginx web server**
- Configured it to serve the portfolio

👉 Nginx acts as:

- Web server
- Reverse proxy

🔷 **7. Deployed Website to VM**

👉 Temporary external IP: http://35.222.34.55/

🔷 **8. Reserved Static IP**

- Converted temporary external IP → **Static IP**

👉 Important because:

- Domain requires a **stable IP target**
- Prevents downtime after VM restart

🔷 **9. Purchased Domain**

- Bought domain: manojpanduraj.com

🔷 **10. Configured DNS**

- Added **A records** pointing domain → VM static IP

👉 Enabled:

- manojpanduraj.com

- www.manojpanduraj.com

🔷 11. Debugged DNS Issue

- Verifying DNS records

🔷 12. Enabled HTTPS (Production Security)

- Installed SSL using **Let's Encrypt (Certbot)**
- Configured HTTPS + auto-renewal

🎯 Final Production Result

https://manojpanduraj.com

- Fully deployed on GCP VM
- HTTPS enabled 🔒
- Domain connected
- Nginx serving content

🌍 Environment Flow (Important)

👉 Local → Dev
👉 Firebase → UAT
👉 GCP VM → Production

👉 This is a **multi-environment deployment approach**